PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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Abstract

This pilot study examines whether inter-individual differences moderate susceptibility to cyber-manipulation in a high-dimensional political opinion state space. We implemented an ontology-driven multi-agent simulation pipeline linking PROFILE, ATTACK, and OPINION ontologies, generated baseline and post-exposure opinions with structured large-language-model agents, and estimated moderation with a baseline-adjusted structural equation model supplemented by robust regression and bootstrap intervals. Run run_4 generated 48 scenarios (24 treated; 24 control) across 6 repeated opinion leaves in Immigration_and_Citizenship. The treated condition showed a mean opinion delta of 42.9 versus 4.1 in control scenarios. In the primary robust model, the attack-by-susceptibility interaction was estimated at -1.06 ($p = 0.875$). The pipeline also produced auditable realism and coherence checks, interactive inspection outputs, and print-ready publication assets. Findings are exploratory and should be read as methodological validation rather than causal evidence about real-world political persuasion.

1 Introduction

Digitally networked political persuasion increasingly unfolds through disinformation, algorithmically amplified attention, and socially embedded manipulation rather than through overtly technical intrusion (Lazer et al., 2018; Vosoughi et al., 2018; Bennett and Livingston, 2018). False or misleading political content can diffuse rapidly online, while citizens' truth discernment is shaped by reasoning style, attention, familiarity, and platform design (Lewandowsky et al., 2017; Pennycook and Rand, 2019, 2021). These processes matter not only for misinformation research in the narrow sense, but also for the broader geopolitical problem of how democratic publics maintain autonomy under persistent informational pressure.

Within security studies, these dynamics are increasingly interpreted as problems of cognitive warfare or cognitive security because the strategic target is not merely information accuracy but the shaping of attention, belief revision, affect, and institutional trust at population scale (Hung and Hung, 2022; Miller, 2023). NATO-linked concept development has gone further by describing the cognitive domain as a new space of competition beyond the traditional land,

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maritime, air, cyber, and space domains, and by emphasizing cognitive superiority as a strategic requirement for contemporary conflict and competition (Atkinson, 2021; Paulauskas, 2024). This framing is useful for the present study because it foregrounds the political and societal consequences of sustained influence operations without collapsing them into purely technical notions of cyber attack. It also clarifies why susceptibility is unlikely to be homogeneous across individuals: the same adversarial stimulus can be filtered differently depending on prior beliefs, trait structure, and issue-specific psychological salience.

Recent work suggests that digital persuasion can be made more effective when messages are tailored to psychological characteristics (Matz et al., 2017), while resilience can be strengthened through cognitive tools, accuracy-oriented interventions, and inoculation strategies (Kozyreva et al., 2020; Pennycook et al., 2021; Roozenbeek et al., 2022). In this project, we use the term *cognitive sovereignty* as an analytic framing for the extent to which a person's political judgement remains self-directed under adversarial informational pressure. We use the term as a methodological synthesis rather than as a settled psychometric construct: it denotes preserved judgemental autonomy under digitally mediated attempts at manipulation.

Direct empirical estimation of PROFILE $\times$ ATTACK $\rightarrow$ OPINION moderation is difficult because suitable datasets are often inaccessible, ethically sensitive, poorly aligned with ontology-driven constructs, or unable to recover comparable baseline and post-exposure states for the same individual. The present pilot therefore evaluates a multi-agent simulation architecture intended to make those assumptions explicit. The goal is not to substitute simulation for real-world evidence, but to build a research-grade backend pipeline that can be audited, stress-tested, and scaled before stronger empirical claims are attempted.

2 Materials and Methods

2.1 Study Design

Run run_4 used the test ontology triplet and generated 48 scenarios with a 50/50 control-treatment split. The treatment arm used a single common misinformation vector, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`, to constrain attack-side variance while the pilot methodology was being tightened. Scenario creation used a repeated-leaf blocked design rather than near-unique issue sampling: 6 opinion leaves from Immigration_and_Citizenship were repeated across the sample, and candidate profiles were oversampled before selection so that low- and high-susceptibility cases were represented across both conditions. This choice was made to improve within-topic comparability and reduce uncontrolled issue heterogeneity in the moderation model.

PROFILE configurations included mixed variable types, combining categorical attributes such as sex with continuous percentile-based traits derived from Big Five facets (Costa and McCrae, 1992). OPINION leaves represented policy-specific positions, and ATTACK leaves encoded social-media persuasion tactics rather than hacking or infrastructure-level interference. Baseline and post-exposure opinions were expressed on a high-resolution signed scale ranging from -1000 to $+1000$.

2.2 Agentic Pipeline

The pipeline proceeded in sequential stages: scenario creation, baseline opinion assessment, attack generation, post-attack opinion assessment, delta construction, moderation modeling,

interactive visualization, publication-asset generation, and manuscript build. Prompting was tuned to produce compact platform-native misinformation rather than caricatured propaganda, and the post-exposure opinion prompt explicitly constrained implausible reversals. Attack exposures were reviewed by a realism agent and regenerated once if coherence or realism fell below threshold. Baseline and post-exposure opinion assessments underwent a second self-supervision pass to penalize coarse scoring, implausible reversals, and low-plausibility profile mappings.

2.3 Modeling Strategy

The primary moderator was fixed in advance as `profile_cont_susceptibility_index`. To avoid algebraic redundancy between `delta_score` and `post_score`, the main structural equation model treated baseline opinion as a pre-exposure anchor and post-exposure opinion as the primary outcome. The SEM followed a parsimonious baseline-adjusted pilot specification inspired by standard SEM practice (Rosseel, 2012):

$$\text{baseline_score}_i = \alpha_b + \beta_1 \text{Susceptibility}_i + \gamma' \text{LeafFE}_i + \varepsilon_{bi} \tag{1}$$

$$\text{post_score}_i = \alpha_p + \lambda \text{baseline_score}_i + \beta_2 \text{Attack}_i + \beta_3 \text{Susceptibility}_i \tag{2}$$

$$+ \beta_4 \left(\text{Attack}_i \times \text{Susceptibility}_i\right) + \delta' \text{LeafFE}_i + \varepsilon_{pi} \tag{3}$$

The opinion delta (`post_score - baseline_score`) was retained as a descriptive quantity and visualization target, but not modeled as a second endogenous SEM variable. Robust OLS estimates with the same right-hand side as the primary post-exposure model and non-parametric bootstrap intervals (Efron and Tibshirani, 1993) were reported alongside the SEM.

3 Results

Figure 1 summarizes the workflow, Figure 2 shows the treated-control delta distribution, Figure 3 visualizes the primary moderation trend, and Figure 4 summarizes primary and exploratory interaction coefficients.

Figure 1: Figure 1. Pilot study design and pipeline schematic.

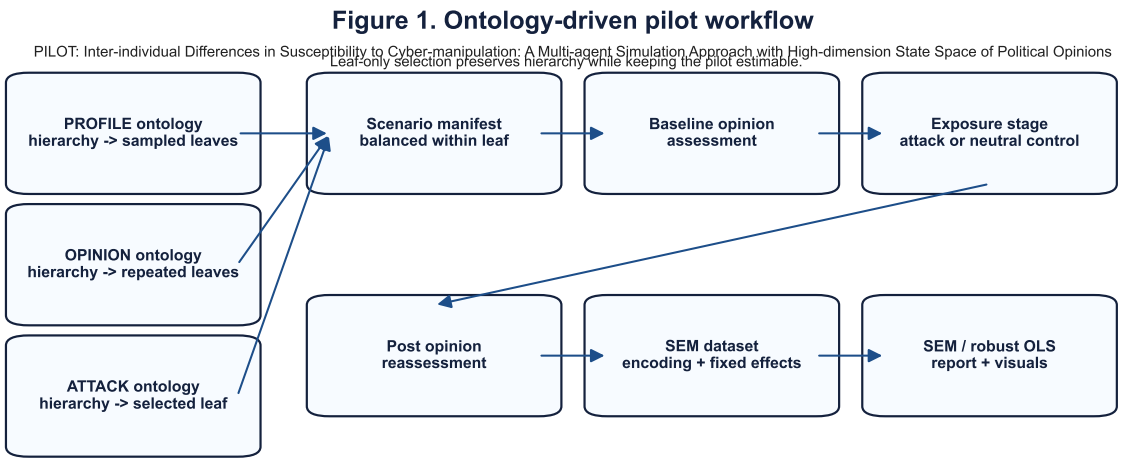

Figure 2: Figure 2. Attack versus control delta distribution with raw scenario points.

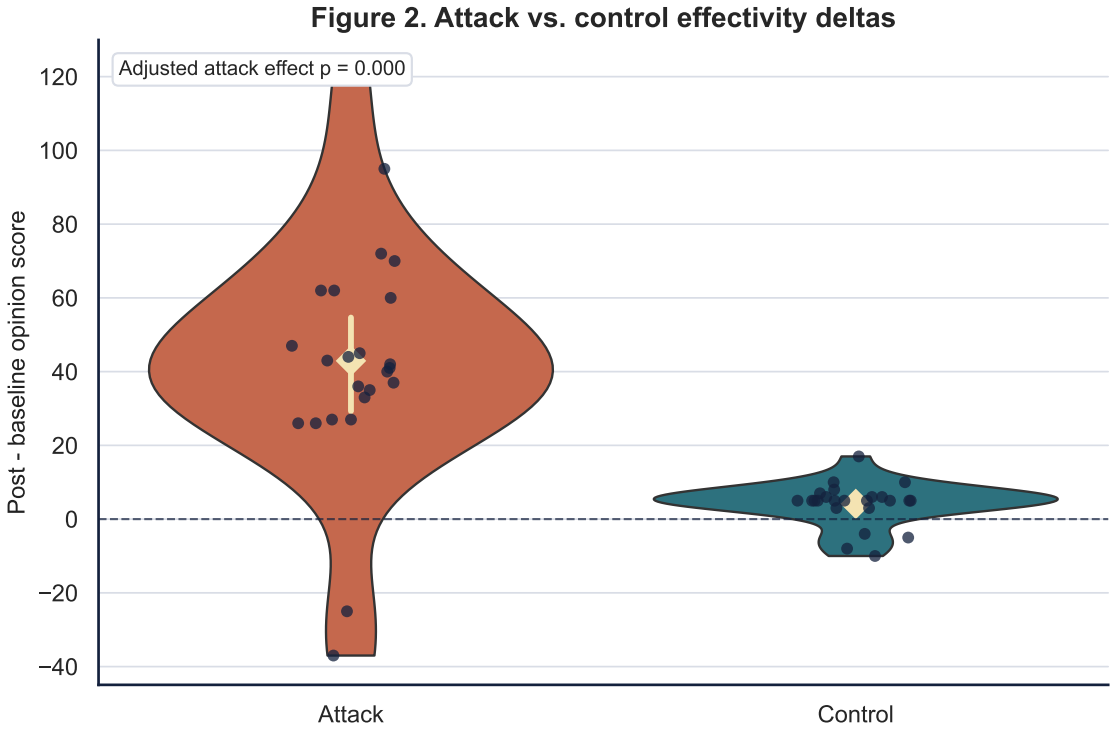

Figure 3: Figure 3. Primary moderation interaction over susceptibility.

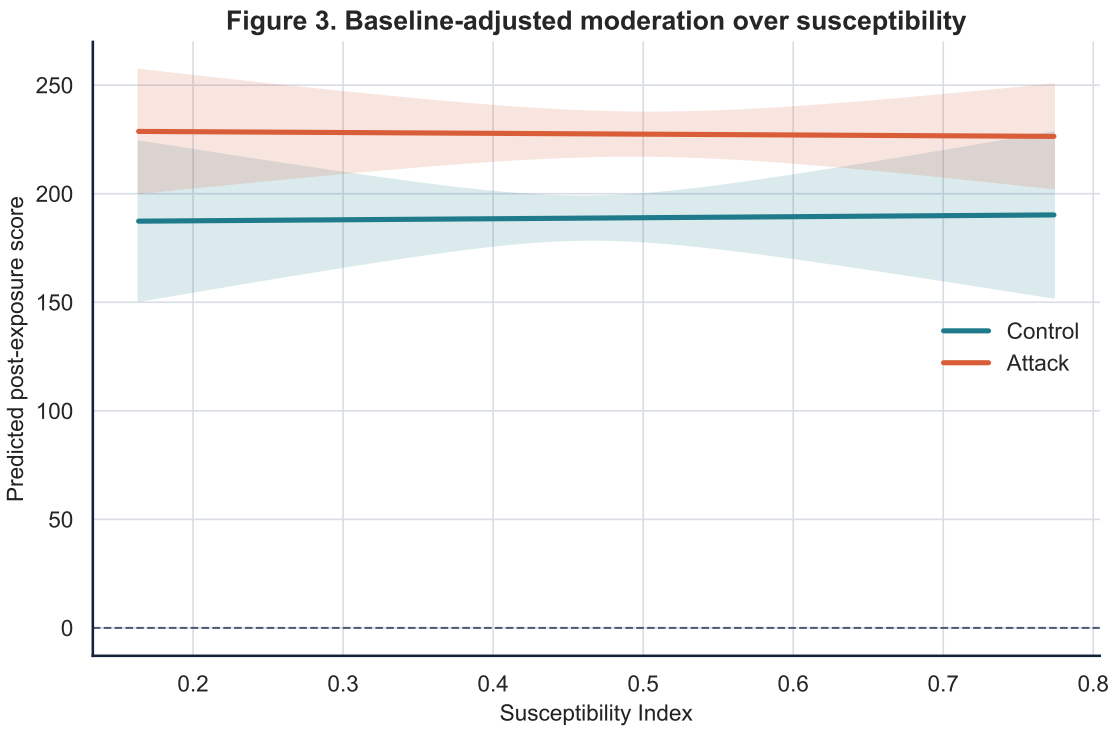

Figure 4: Figure 4. Primary and exploratory moderator interaction coefficients.

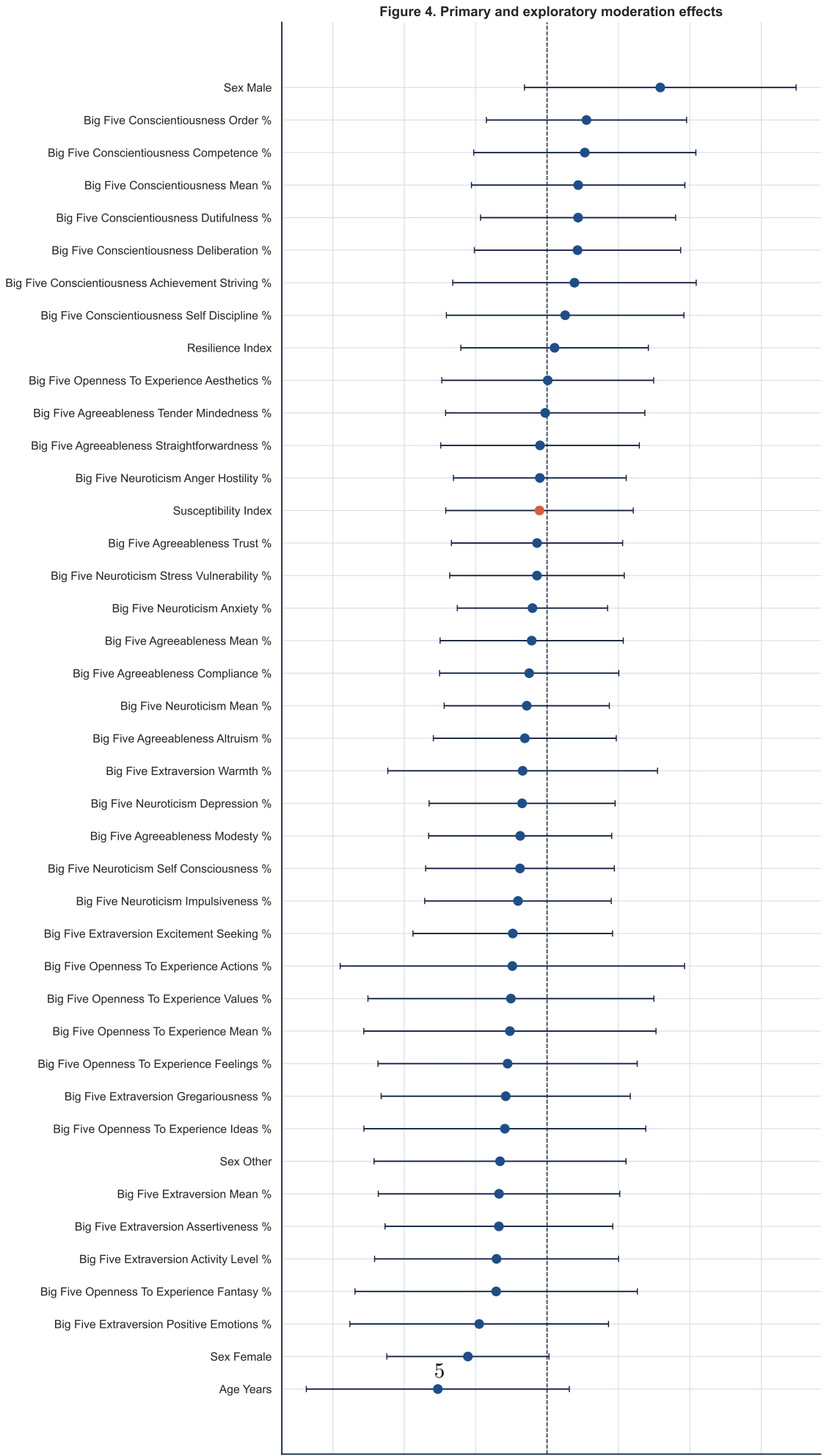

Table 1: Table 1. Pilot design and run configuration

Parameter	Value
Run identifier	run_4
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Scenarios	48
Attack ratio	0.500
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
OpenRouter model	mistralai/mistral-small-3.2-24b-instruct
Primary moderator	profile_cont_susceptibility_index
Bootstrap samples	800
Self-supervised attack realism	True
Test ontology	True

Note. Run-level configuration for the pilot simulation reported in run_4.

Table 2: Table 2. Descriptive statistics by condition

Condition	n	baseline_mean	post_mean	delta_mean	delta_sd	susceptibility_mean	realism_mean
Attack	24	167.750	210.667	42.917	32.252	0.505	0.792
Control	24	201.375	205.500	4.125	5.803	0.460	0.000

Note. Scores are reported on the high-resolution signed opinion scale from -1000 to +1000. Realism scores apply only to treated scenarios.

Table 3: Table 3. Primary baseline-adjusted moderation model with robust and bootstrapped estimates

term	estimate	std_error	p_value	conf_low_x	conf_high_x	bootstrap_mean	bootstrap_std	conf_low_y	conf_high_y	n_bootstrap_success
Intercept	15.149	12.183	0.214	-8.730	39.028	15.150	12.255	-4.737	42.270	800
baseline_score	0.996	0.022	0.000	0.953	1.039	0.997	0.024	0.948	1.044	800
attack_present	38.641	7.655	0.000	23.637	53.645	38.515	7.615	23.019	52.664	800
primary_moderator_z	0.589	3.360	0.861	-5.997	7.175	0.559	4.713	-7.747	9.819	800
attack_x_primary_moderator	-1.056	6.701	0.875	-14.189	12.078	-0.921	7.159	-15.696	13.099	800
opinion_leaf_fe_issue_position_taxonomy____immigration_and_citizenship____border_enforcement_expansion_support	-14.058	14.184	0.322	-41.858	13.742	-14.215	13.534	-45.295	8.936	800
opinion_leaf_fe_issue_position_taxonomy____immigration_and_citizenship____detention_of_irregular_migrants_acceptance	-15.477	17.863	0.386	-50.488	19.533	-16.318	17.202	-56.469	12.047	800
opinion_leaf_fe_issue_position_taxonomy____immigration_and_citizenship____immigration_level_increase_support	-8.188	14.174	0.564	-35.968	19.593	-8.243	13.014	-35.185	13.099	800
opinion_leaf_fe_issue_position_taxonomy____immigration_and_citizenship____multiculturalism_support	-22.240	17.056	0.192	-55.669	11.190	-22.381	15.906	-58.091	3.443	800
opinion_leaf_fe_issue_position_taxonomy____immigration_and_citizenship____temporary_work_visas_expansion_support	-0.924	17.375	0.958	-34.978	33.129	-1.105	15.398	-33.854	26.173	800

Note. The primary outcome is post-exposure opinion score with baseline opinion and opinion-leaf fixed effects included as controls. Robust standard errors are HC3. Bootstrap intervals are percentile intervals from resampled pilot fits and are exploratory given the small sample.

The attack condition shifted opinions more strongly than the control condition in directional terms (mean treated delta = 42.9; mean control delta = 4.1). In the primary robust baseline-adjusted post model, the attack coefficient was 38.64 with p = 0.000, while the primary moderation coefficient was -1.06 with p = 0.875. Baseline opinion remained the dominant stabilizing predictor (robust estimate = 0.996, p = 0.000). Bootstrap interval for the primary interaction was unavailable.

The SEM converged. SEM fit was CFI = 0.720 and RMSEA = 0.232. Treated exposures achieved a mean realism score of 0.79, indicating that the self-supervised reviewer generally accepted the generated persuasion content as platform-plausible. The main post-exposure equation followed the baseline-adjusted specification shown in the Methods section, while the

supplementary SEM overview annotates the central path coefficients directly. The most informative exploratory moderators were Sex Female (interaction est. -11.08, p = 0.056); Sex Male (interaction est. 15.86, p = 0.102); Age Years (interaction est. -15.29, p = 0.104).

4 Discussion

The present pilot demonstrates that the pipeline works end to end: ontology-guided scenario generation, structured LLM assessments, realism/coherence review loops, moderation estimation, static figure export, interactive SEM inspection, and automated manuscript generation all completed within a single reproducible workflow. This remains the primary result of the pilot.

Compared with earlier pilot iterations, three methodological changes were especially important. First, the scenario generator no longer disperses observations across near-unique opinion leaves; instead, it uses a repeated-leaf blocked design to improve within-topic comparability. Second, the primary SEM no longer models both `delta_score` and `post_score` as endogenous outcomes despite their algebraic dependence. Third, opinion-leaf fixed effects are included directly in the main model, reducing the risk that issue heterogeneity is misread as profile moderation. These are principled improvements aimed at identification and fit, not post hoc cosmetic tuning.

At the same time, inferential limitations remain substantial. With only 48 scenarios, even an improved pilot design cannot support strong substantive claims about real-world susceptibility. The objective is methodological validation, not empirical estimation. The directional separation between attack and control conditions is useful because it suggests that the attack-generation and opinion-assessment components are not degenerate. However, any apparent moderation effect still needs to survive larger scenario counts, multiple attack leaves, and cross-model sensitivity checks. Even after the present redesign, conventional SEM fit benchmarks are not yet met; the remaining misfit is plausibly driven by small-N estimation, residual LLM response variability, and the burden of issue fixed effects relative to sample size.

Three additional features improve the credibility of the pilot. First, the primary moderator was fixed in advance rather than selected opportunistically from whichever variable produced the strongest coefficient. Second, attack realism and opinion coherence were explicitly audited with repair loops. Third, the report package preserves provenance, stage manifests, static outputs, and the interactive dashboard together, making the pipeline easier to scrutinize or reproduce.

5 Conclusion

This pilot supports the feasibility of an ontology-driven multi-agent simulation framework for studying how inter-individual differences may moderate susceptibility to cyber-manipulation in political opinion spaces. The pipeline is operational, auditable, and sufficiently modular to scale beyond the present pilot. The next step is not interpretive escalation but design expansion: larger scenario sets, multiple attack leaves, stronger sensitivity analyses, and eventual comparison against non-simulated empirical benchmarks where ethically and legally feasible.

A Supplementary Materials

The repository also includes an interactive SEM dashboard generated from run run_4. The static publication assets below are the paper-ready subset of that richer exploratory output.

Figure 5: Supplementary Figure S1. Baseline versus post-attack opinion scores.

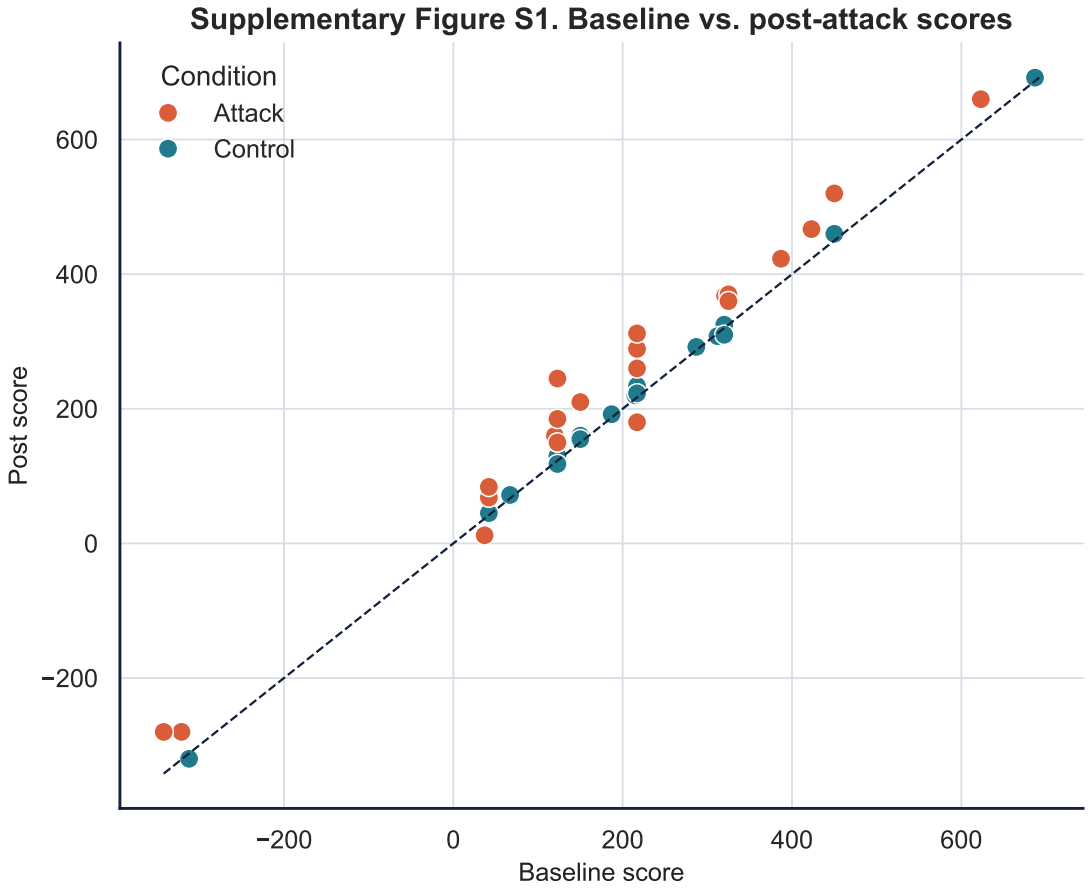

Figure 6: Supplementary Figure S2. Attack realism and coherence distributions.

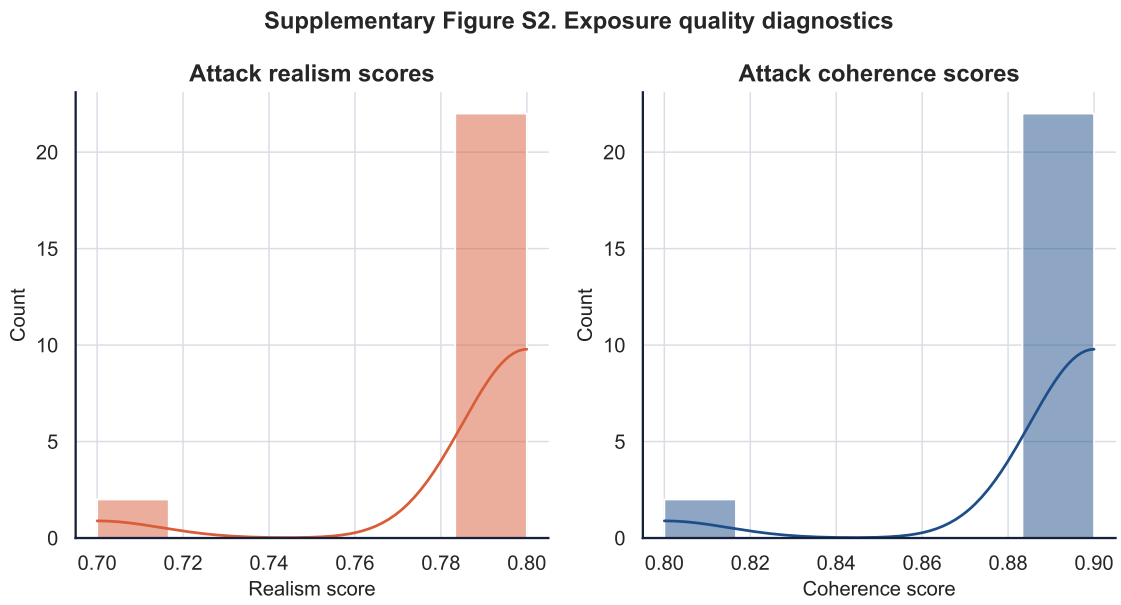

Figure 7: Supplementary Figure S3. Scenario composition across opinion leaves.

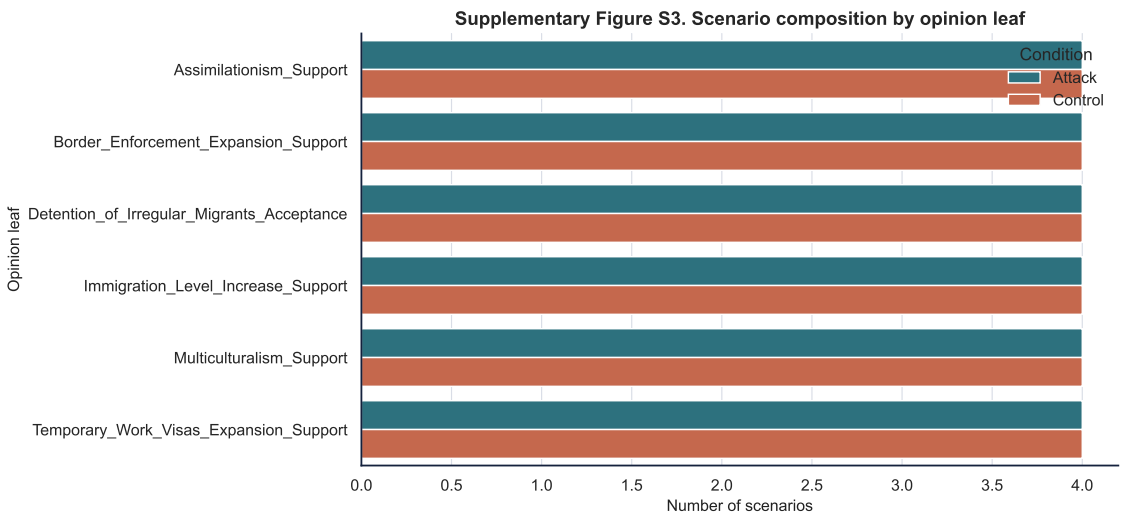

Figure 8: Supplementary Figure S4. Primary moderation model overview.

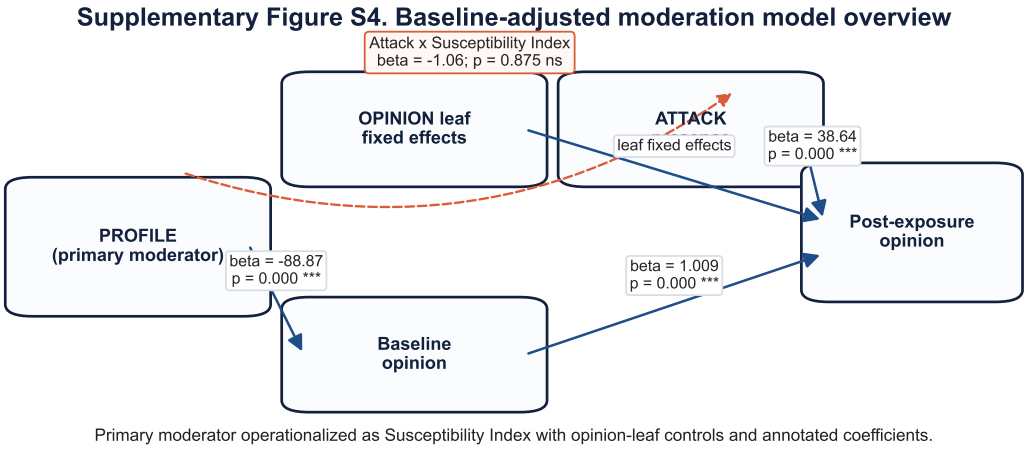

Table 4: Supplementary Table S1. Ontology leaves used in the pilot

ontology	leaf_label	leaf_path	n_scenarios
OPINION	Assimilationism_Support	Issue_Position_Taxonomy > Immigration_and_Citizenship > Assimilationism_Support	8
OPINION	Border_Enforcement_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citizenship > Border_Enforcement_Expansion_Support	8
OPINION	Detention_of_Irregular_Migrants_Acceptance	Issue_Position_Taxonomy > Immigration_and_Citizenship > Detention_of_Irregular_Migrants_Acceptance	8
OPINION	Immigration_Level_Increase_Support	Issue_Position_Taxonomy > Immigration_and_Citizenship > Immigration_Level_Increase_Support	8
OPINION	Multiculturalism_Support	Issue_Position_Taxonomy > Immigration_and_Citizenship > Multiculturalism_Support	8
OPINION	Temporary_Work_Visas_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citizenship > Temporary_Work_Visas_Expansion_Support	8
ATTACK	Misleading narrative framing	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing	24

Note. Opinion leaves reflect the scenario allocation in run_4; the attack ontology was restricted to one common misinformation vector for this pilot.

Table 5: Supplementary Table S2. Exploratory moderator comparison

moderator_column	moderator_label	role	interaction_estimate	interaction_std_error	interaction_p_value	interaction_conf_low	interaction_conf_high	moderator_main_estimate	moderator_main_p_value
profile_cat__profile_cat_sex_Female	Sex Female	exploratory	-11.081	5.794	0.056	-22.437	0.275	4.647	0.225
profile_cat__profile_cat_sex_Male	Sex Male	exploratory	15.860	9.706	0.102	-3.163	34.884	-6.029	0.121
profile_cont_age_years	Age Years	exploratory	-15.294	9.396	0.104	-33.710	3.123	-0.583	0.795
profile_cont_big_five_extraversion_positive_emotions_pct	Big Five Extraversion Positive Emotions %	exploratory	-9.501	9.240	0.304	-27.610	8.609	-1.031	0.695
profile_cont_big_five_extraversion_assertiveness_pct	Big Five Extraversion Assertiveness %	exploratory	-6.743	8.137	0.407	-22.690	9.205	-0.843	0.791
profile_cont_big_five_extraversion_activity_level_pct	Big Five Extraversion Activity Level %	exploratory	-7.071	8.715	0.417	-24.151	10.010	-1.375	0.569
profile_cont_big_five_extraversion_mean_pct	Big Five Extraversion Mean %	exploratory	-6.722	8.628	0.436	-23.633	10.188	-1.141	0.682
profile_cont_big_five_conscientiousness_order_pct	Big Five Conscientiousness Order %	exploratory	5.525	7.153	0.440	-8.496	19.545	1.734	0.591
profile_cat__profile_cat_sex_Other	Sex Other	exploratory	-6.581	9.002	0.465	-24.225	11.063	1.842	0.623
profile_cont_big_five_openness_to_experience_fantasy_pct	Big Five Openness To Experience Fantasy %	exploratory	-7.125	10.089	0.480	-26.900	12.650	0.703	0.805
profile_cont_big_five_extraversion_excitement_seeking_pct	Big Five Extraversion Excitement Seeking %	exploratory	-4.801	7.130	0.501	-18.776	9.174	-0.542	0.845
profile_cont_big_five_conscientiousness_competence_pct	Big Five Conscientiousness Competence %	exploratory	5.288	7.936	0.505	-10.265	20.842	2.408	0.587
profile_cont_big_five_extraversion_gregariousness_pct	Big Five Extraversion Gregariousness %	exploratory	-5.784	8.900	0.516	-23.227	11.659	-0.658	0.812
profile_cont_big_five_conscientiousness_dutifulness_pct	Big Five Conscientiousness Dutifulness %	exploratory	4.345	6.970	0.533	-9.317	18.007	2.297	0.468
profile_cont_big_five_neuroticism_impulsiveness_pct	Big Five Neuroticism Impulsiveness %	exploratory	-4.068	6.664	0.542	-17.129	8.993	2.692	0.508
profile_cont_big_five_openness_to_experience_feelings_pct	Big Five Openness To Experience Feelings %	exploratory	-5.522	9.263	0.551	-23.676	12.633	1.209	0.726
profile_cont_big_five_openness_to_experience_ideas_pct	Big Five Openness To Experience Ideas %	exploratory	-5.912	10.071	0.557	-25.649	13.826	0.927	0.757
profile_cont_big_five_conscientiousness_deliberation_pct	Big Five Conscientiousness Deliberation %	exploratory	4.268	7.370	0.563	-10.177	18.713	1.403	0.680
profile_cont_big_five_agreeableness_modesty_pct	Big Five Agreeableness Modesty %	exploratory	-3.762	6.541	0.565	-16.581	9.058	1.152	0.775
profile_cont_big_five_conscientiousness_mean_pct	Big Five Conscientiousness Mean %	exploratory	4.362	7.625	0.567	-10.583	19.307	1.940	0.607
profile_cont_big_five_neuroticism_self_consciousness_pct	Big Five Neuroticism Self Consciousness %	exploratory	-3.789	6.740	0.574	-17.000	9.422	0.499	0.896
profile_cont_big_five_neuroticism_depression_pct	Big Five Neuroticism Depression %	exploratory	-3.492	6.645	0.599	-16.516	9.533	-1.360	0.727
profile_cont_big_five_openness_to_experience_mean_pct	Big Five Openness To Experience Mean %	exploratory	-5.200	10.439	0.618	-25.659	15.260	0.718	0.806
profile_cont_big_five_openness_to_experience_values_pct	Big Five Openness To Experience Values %	exploratory	-5.061	10.214	0.620	-25.080	14.959	-0.115	0.970
profile_cont_big_five_neuroticism_mean_pct	Big Five Neuroticism Mean %	exploratory	-2.840	5.900	0.630	-14.404	8.723	0.931	0.776
profile_cont_big_five_agreeableness_altruism_pct	Big Five Agreeableness Altruism %	exploratory	-3.112	6.536	0.634	-15.922	9.697	1.211	0.756
profile_cont_big_five_conscientiousness_achievement_striving_pct	Big Five Conscientiousness Achievement Striving %	exploratory	3.841	8.698	0.659	-13.207	20.890	0.633	0.863
profile_cont_big_five_openness_to_experience_actions_pct	Big Five Openness To Experience Actions %	exploratory	-4.857	12.302	0.693	-28.968	19.254	0.807	0.747
profile_cont_big_five_agreeableness_compliance_pct	Big Five Agreeableness Compliance %	exploratory	-2.501	6.405	0.696	-15.054	10.052	1.423	0.721
profile_cont_big_five_neuroticism_anxiety_pct	Big Five Neuroticism Anxiety %	exploratory	-2.043	5.370	0.704	-12.568	8.481	1.192	0.690
profile_cont_big_five_extraversion_warmth_pct	Big Five Extraversion Warmth %	exploratory	-3.417	9.638	0.723	-22.397	15.473	-1.601	0.616
profile_cont_big_five_agreeableness_mean_pct	Big Five Agreeableness Mean %	exploratory	-2.152	6.545	0.742	-14.980	10.677	0.662	0.868
profile_cont_big_five_conscientiousness_self_discipline_pct	Big Five Conscientiousness Self Discipline %	exploratory	2.538	8.488	0.765	-14.098	19.174	1.679	0.701
profile_cont_big_five_agreeableness_trust_pct	Big Five Agreeableness Trust %	exploratory	-1.400	6.116	0.819	-13.388	10.587	0.641	0.885
profile_cont_big_five_neuroticism_stress_vulnerability_pct	Big Five Neuroticism Stress Vulnerability %	exploratory	-1.411	6.236	0.821	-13.635	10.812	0.955	0.721
profile_cont_big_five_neuroticism_anger_hostility_pct	Big Five Neuroticism Anger Hostility %	exploratory	-1.008	6.174	0.870	-13.108	11.092	1.183	0.714
profile_cont_resilience_index	Resilience Index	exploratory	1.056	6.701	0.875	-12.078	14.189	-0.589	0.861
profile_cont_big_five_agreeableness_straightforwardness_pct	Big Five Agreeableness Straightforwardness %	exploratory	-0.981	7.097	0.890	-14.890	12.928	-0.129	0.973
profile_cont_big_five_agreeableness_tender_mindedness_pct	Big Five Agreeableness Tender Mindedness %	exploratory	-0.255	7.118	0.971	-14.205	13.696	-0.819	0.795
profile_cont_big_five_openness_to_experience_aesthetics_pct	Big Five Openness To Experience Aesthetics %	exploratory	0.098	7.568	0.990	-14.735	14.930	0.348	0.915
profile_cont_susceptibility_index	Susceptibility Index	primary	-1.056	6.701	0.875	-14.189	12.078	0.589	0.861

Note. Each moderator was estimated in a separate attack-by-moderator interaction model with baseline opinion included as a covariate.

References

Atkinson, E. (2021). Countering cognitive warfare: Awareness and resilience. https://www.nato.int/docu/review/articles/2021/05/20/

Table 6: Supplementary Table S3. Assumption and peer-review risk register

section	item	status	detail
Assumption register	Pilot sample size is sufficient for stable moderation estimates.	risk	Treat estimates as exploratory and scale scenario count in later runs.
Assumption register	Treatment-control balance is adequate.	ok	Use explicit stratified assignment in scenario generator.
Assumption register	Opinion measurement has enough numeric resolution.	ok	Prompt enforces high-resolution scores in [-1000,1000].
Assumption register	Attack text realism is adequate.	ok	Self-supervised realism reviewer + rewrite loop in Stage 03.
Assumption register	SEM converges with acceptable fit.	risk	Use robust OLS supplement, reduce heterogeneity, and increase scenario count before substantive interpretation.
Assumption register	LLM outputs are reproducible and auditable.	ok	Persist all prompts/outputs in provenance/raw_llm.
Peer-review critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addressed	Added transparent assumption register and explicit exploratory caveat in reports.
Peer-review critique	Adversarial content may be unrealistic or too generic.	addressed	Added realism review agent with rewrite loop and heuristic checks.
Peer-review critique	Model dependence on a single LLM could bias outputs.	addressed	Model is CLI-configurable and run metadata captures exact model for replication.
Peer-review critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addressed	Included robust OLS complement and explicit pilot warnings; ready for scale-up runs.
Peer-review critique	Global SEM fit remains below conventional thresholds despite convergence.	addressed	Blocked opinion-leaf design, baseline-adjusted post model, fixed effects, and robust OLS/bootstraps were added; further scale-up is still required.

Note. This table summarizes the principal methodological assumptions and the explicit mitigations implemented in the pilot pipeline.

Table 7: Supplementary Table S4. Reproducibility manifest

Field	Value
run_id	run_4
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalD
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalD
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
primary_moderator	profile_cont_susceptibility_index
opinion_leaf_count	38
attack_leaf_count	9

Note. Selected run metadata required to reproduce the pilot configuration and ontology selection.

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